

The Rank Illusion: Does NA Gold Match Korean Bronze?

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Abstract

This paper presents an analysis of the popular e-sports game League of Legends, specifically comparing North American Gold-ranked players to Korean Bronze-ranked players. By examining the average in-game economy per region and key gameplay aspects, this study aims to determine which rank demonstrates a higher overall skill level.

1. Introduction

League of Legends has been a popular game around the world dominating the e-sports scene. A year after releasing in 2009, the Monthly Active Users (MAU) was 15 million players. Comparing the MAU today, it is around 140 to 160 million players worldwide, according to Riot.

LoL's World championship is one of the biggest e-sports tournament at the end of the year, where each region's best teams goes against each other to be crowned the best of that year. The LCK (League Championship Korea) , has one 9 Worlds championship out of the 15 years that Worlds has been a tournament, establishing themselves as a top tier in LoL. The LCS (League Championship Series) or North America has never won worlds or has never been to the final stage of the tournament.

Looking at video game culture, Korea has a stronger esports scene, deeper social integration of gaming, and more professional opportunities compared to North America. As Korea and South East Asia have the integration of Internet Cafe's where players can rent out computers to play games with their friends in person. Where in America, most players have their own set ups in their homes.

2. Data Collection

The data collected was provided by Riot Games' API service. Initially, a random set of players from both servers was selected, and a random set of games per user was collected to account for tilt queuing. Approximately looking at 8,000 games and in one game there are 10 players. The data is from Patch 15 of League of Legends.

3. Economy

The economy of league of legends revolves around gold, which is the in-game currency used to purchase items that

enhance a champion's ability to affect a game. Gold is earned through various ways, including

- Last-hitting Minions – The primary and most consistent way to generate gold.
- Champion Kills & Assists – Grant bonus gold based on enemy kill streaks.
- Turret & Objective Rewards – Destroying turrets, securing dragons, Baron Nashor, or Rift Herald provides team-wide gold.
- Jungle Camps – Killing jungle monsters rewards gold, crucial for junglers.
- Passive Income – A small amount of gold is gained over time.
- Support Items & Bounties – Certain items grant gold through objectives or champion interactions, and bounty systems reward teams for taking down high-value targets.

North America - Blue | Korea - Red

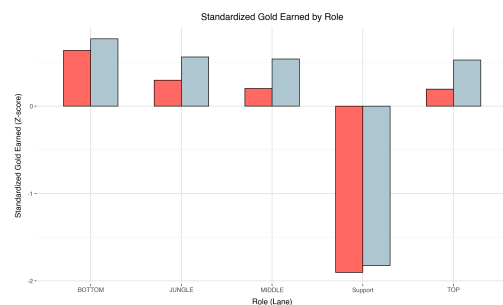


Figure 1: Gold Earned

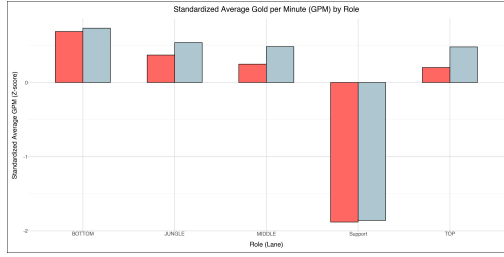


Figure 2: Gold Per Minute

North American Gold ranked players earn more gold and have collect more gold per minute compared to their Bronze ranked Korean counter parts. This shows that, North American players are more focused on snowballing and farming compared to more macro plays that Korea is famous for. Snowballing is the term used to explain a player or team that is taking every small advantage and building on it. Think about rolling a snowball down a hill and getting large at the end.

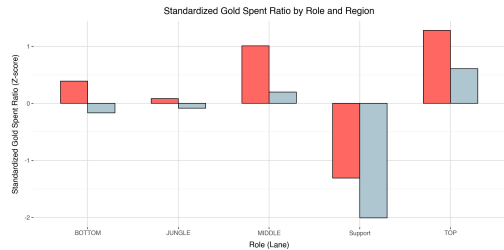


Figure 3: Gold Spent Ratio

Even though North America earns more gold overall over the course of the game, they are just sitting on that gold and mindlessly farming. Korean players on the other hand are spending their gold to get items for their champions which leads to big advantages. Accumulating gold is great, but spending it and making impact is much for significant.

4. Gameplay

League of Legends gameplay is complex, with many dimensions to consider. When evaluating players, we will focus on their individual mechanics and overall impact on the game, particularly in terms of dealing damage to objectives.

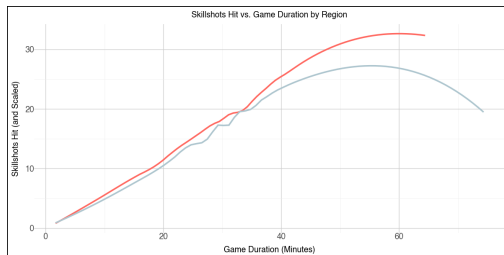


Figure 4: Skill Shots Hit

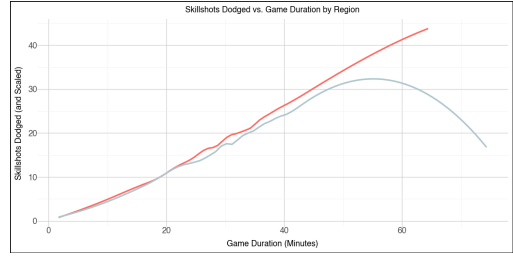


Figure 5: Skill Shots Dodged

A skill shot is an ability that requires some level of aim. It is not a point and click ability but an ability of a champion that requires a little bit more skill.

Looking at the skill shots hit, Korean players in Bronze are hitting more of their skill shots compared to North America. As the games goes on, Korean players are casting abilities that are going to hit compared to the spam that North America is doing.

This also reflects the skill shots dodged. This column in the dataset is counted by a champion who dodges another players or champion skill shot when in proximity. Korean players are more skilled when they are dodging skill shots even in the late game. American players in the late game show a trend of not dodging abilities.

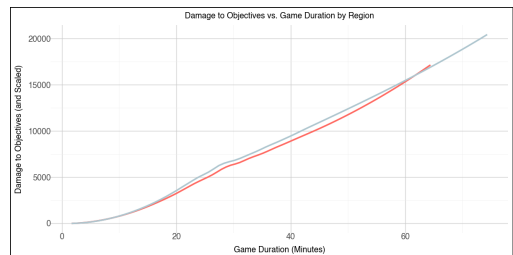


Figure 6: Damage to Objectives

There is a strong correlation between objective damage and game length. In North America, Gold-ranked games tend to last longer on average compared to Bronze-ranked games. This highlights the efficiency of Korean players, even in Bronze!

5. Conclusion

The joke that North American Gold is worse than Korean Bronze seems to hold some truth, much to the disappointment of North American fans. When comparing the in-game economy of League of Legends, North American Gold players appear to earn gold at a higher rate, but Korean players utilize their resources more efficiently to close out games faster. From a gameplay perspective, it's clear that Korean players demonstrate superior skill, landing more abilities and dodging more effectively.